

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET F — RAPID FIRE SPEED TEST (30 MINS)****CHAPTER 03: ATMOSPHERE AND CLIMATE**

TIME ALLOWED: 30 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-F-03

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET F - Rapid Fire Speed Test):

1. The question paper comprises **25 Rapid Fire Multiple Choice Questions** of **1 mark each**.
2. Designed for **Rapid Recall & High-Speed Accuracy**: Recommended target time is **30 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

⚡ SPEED CHALLENGE INSTRUCTIONS: Answer promptly without looking at notes. Test your instant recall of historical milestones, geographic terms, constitutional provisions, and core economic laws!

Q.1 In which atmospheric layer do virtually all convective weather phenomena (clouds, storms, rain) occur?

- (A) Stratosphere (B) Troposphere
(C) Mesosphere (D) Thermosphere

Q.2 What is the normal environmental lapse rate in the troposphere per 1,000 metres of elevation?

- (A) 1°C per 1,000 m (B) 6.5°C per 1,000 m
(C) 15°C per 1,000 m (D) 0°C per 1,000 m

Q.3 Which atmospheric layer contains the maximum concentration of the protective ozone (O₃) layer?

- (A) Troposphere (B) Stratosphere
(C) Mesosphere (D) Exosphere

Q.4 What is the coldest atmospheric layer, where meteors burn up upon entering Earth's atmosphere?

- (A) Troposphere (B) Stratosphere
(C) Mesosphere (D) Thermosphere

Q.5 Which gas is the most abundant constituent in the Earth's dry atmosphere by volume?

- (A) Oxygen (~21%) (B) Nitrogen (~78%)
(C) Carbon dioxide (~0.04%) (D) Argon (~0.93%)

Q.6 According to Ferrel's Law, the Coriolis force deflects moving winds in the Northern Hemisphere to their:

- (A) Left (B) Right
(C) South (D) Vertical upwards

Q.7 At which geographic latitude is the magnitude of the Coriolis force equal to zero?

- | | |
|------------------------------|------------------------------|
| (A) At the North Pole (90°N) | (B) At the Equator (0°) |
| (C) At 45°N | (D) At the South Pole (90°S) |

Q.8 The equatorial belt of low atmospheric pressure characterized by calm, light variable winds is known as the:

- | | |
|---------------------|------------------------|
| (A) Horse Latitudes | (B) Doldrums (or ITCZ) |
| (C) Subpolar Lows | (D) Polar Highs |

Q.9 The subtropical high-pressure belts situated around 30° North and South latitudes are traditionally termed:

- | | |
|---------------------|---------------------|
| (A) Doldrums | (B) Horse Latitudes |
| (C) Roaring Forties | (D) ITCZ |

Q.10 During which months does the Southwest Monsoon season dominate rainfall across most of India?

- | | |
|--------------------------|-----------------------|
| (A) December to February | (B) June to September |
| (C) October to November | (D) March to May |

Q.11 Which Indian coastal region receives the majority of its annual rainfall during the winter (Oct–Dec) retreating monsoon?

- | | |
|-------------------|-----------------------------------|
| (A) Konkan Coast | (B) Coromandel Coast (Tamil Nadu) |
| (C) Gujarat Coast | (D) Malabar Coast (Kerala) |

Q.12 Which location holds the world record for the highest average annual rainfall on Earth?

- | | |
|--------------------------|--------------------------|
| (A) Jaisalmer, Rajasthan | (B) Mawsynram, Meghalaya |
| (C) Leh, Ladakh | (D) Pune, Maharashtra |

Q.13 Scorching hot, dry westerly winds blowing across the northern plains of India during May and June are called:

- | | |
|-----------------|-------------------|
| (A) Kalbaisakhi | (B) Loo |
| (C) Chinook | (D) Mango Showers |

Q.14 Violent pre-monsoon convective thunderstorms and squalls occurring in West Bengal and Assam are called:

- | | |
|-------------|-------------------------------|
| (A) Loo | (B) Kalbaisakhi (Nor'westers) |
| (C) Mistral | (D) Bora |

Q.15 What minimum ocean sea surface temperature (SST) is required to sustain tropical cyclogenesis?

- | | |
|----------|-----------------------------|
| (A) 0°C | (B) At least 26.5°C to 27°C |
| (C) 15°C | (D) 100°C |

Q.16 The calm, cloudless, low-pressure center of a mature tropical cyclone is called the:

- | | |
|-----------------|------------|
| (A) Eyewall | (B) Eye |
| (C) Spiral band | (D) Trough |

Q.17 Which of the following is the most abundant natural greenhouse gas in the Earth's lower atmosphere?

- | | |
|--------------------------------|-------------------------------------|
| (A) Methane (CH ₄) | (B) Water vapour (H ₂ O) |
| (C) Ozone (O ₃) | (D) Chlorofluorocarbons (CFC) |

Q.18 The fraction of incoming solar radiation reflected back into space by a surface without absorption is termed its:

- (A) Insolation (B) Albedo
(C) Lapse rate (D) Sensible heat

Q.19 Periodic warming of equatorial sea surface temperatures in the eastern Pacific Ocean that often weakens Indian monsoons is:

- (A) La Niña (B) El Niño (ENSO)
(C) Walker Circulation (D) Indian Ocean Dipole (IOD)

Q.20 The atmospheric condition where temperature increases with altitude, trapping pollutants close to the ground, is called:

- (A) Normal lapse rate (B) Temperature inversion (thermal inversion)
(C) Adiabatic expansion (D) Coriolis deflection

Q.21 The international treaty signed in 1987 to phase out ozone-depleting chlorofluorocarbons (CFCs) is the:

- (A) Paris Agreement (B) Montreal Protocol
(C) Kyoto Protocol (D) Geneva Convention

Q.22 Why does coastal Mumbai experience a much milder climate with lower temperature range than inland Delhi?

- (A) Mumbai is located on high mountains (B) The moderating maritime influence of the Arabian Sea due to water's high specific heat capacity
(C) Delhi receives 500 inches of snow every winter (D) Mumbai has zero sunshine year-round

Q.23 Fast-flowing, narrow, meandering air currents in the upper troposphere (altitude 9–14 km) are called:

- (A) Trade winds (B) Jet streams
(C) Sea breezes (D) Monsoonal troughs

Q.24 Pre-monsoon showers in Kerala and coastal Karnataka that aid in the early ripening of mangoes are known as:

- (A) Kalbaisakhi (B) Mango Showers
(C) Loo (D) Cherry Blossom showers

Q.25 The metric quantifying the total greenhouse gases produced directly and indirectly by human activities, measured in \$CO_2e\$, is:

- (A) Carbon footprint (B) Albedo index
(C) Thermal coefficient (D) Specific heat

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET F)

01	(A) (B) (C) (D)	02	(A) (B) (C) (D)	03	(A) (B) (C) (D)	04	(A) (B) (C) (D)	05	(A) (B) (C) (D)
06	(A) (B) (C) (D)	07	(A) (B) (C) (D)	08	(A) (B) (C) (D)	09	(A) (B) (C) (D)	10	(A) (B) (C) (D)
11	(A) (B) (C) (D)	12	(A) (B) (C) (D)	13	(A) (B) (C) (D)	14	(A) (B) (C) (D)	15	(A) (B) (C) (D)
16	(A) (B) (C) (D)	17	(A) (B) (C) (D)	18	(A) (B) (C) (D)	19	(A) (B) (C) (D)	20	(A) (B) (C) (D)
21	(A) (B) (C) (D)	22	(A) (B) (C) (D)	23	(A) (B) (C) (D)	24	(A) (B) (C) (D)	25	(A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & RAPID RECALL EXPLANATIONS (SET F)

Q. No.	Correct	Curricular Concept & Rapid Recall Analysis
Q.1	(B)	The troposphere contains 75% of atmospheric mass and almost all water vapour, hosting all active weather.
Q.2	(B)	The normal lapse rate in the troposphere averages a temperature drop of 6.5°C per kilometre of ascent.
Q.3	(B)	The stratosphere houses the ozone layer (at 15–35 km altitude), which absorbs harmful solar UV rays.
Q.4	(C)	The mesosphere is the coldest layer (plunging below -90°C) and incinerates meteoroids through friction.
Q.5	(B)	Nitrogen constitutes approximately 78.08% of clean dry air by volume.
Q.6	(B)	The Coriolis force deflects moving objects to their right in the Northern Hemisphere and to their left in the Southern Hemisphere.
Q.7	(B)	Coriolis force is proportional to $\sin \phi$; at the equator ($\phi = 0^\circ$), $\sin 0^\circ = 0$, making Coriolis force zero.
Q.8	(B)	The equatorial low-pressure convergence zone is historically known to sailors as the Doldrums.
Q.9	(B)	The calm, dry subtropical high-pressure subsidence belts at 30°–35° latitude are called the Horse Latitudes.
Q.10	(B)	The Southwest Monsoon delivers over 75% of India's annual precipitation between June and September.
Q.11	(B)	The northeast retreating monsoon absorbs moisture over the Bay of Bengal, dropping rain on Tamil Nadu's Coromandel coast.
Q.12	(B)	Mawsynram in the Khasi Hills of Meghalaya receives ~11,872 mm of annual precipitation, making it the wettest place on Earth.
Q.13	(B)	The Loo consists of intensely hot, dry summer winds originating from the Thar desert across North India.
Q.14	(B)	Kalbaisakhi ('calamity of the month of Baisakh') are severe pre-monsoon thunderstorms in Bengal.
Q.15	(B)	Tropical cyclones require sea surface temperatures of at least 26.5°C–27°C to provide sufficient latent heat moisture fuel.
Q.16	(B)	The eye is the central calm zone of subsiding air featuring clear skies and lowest central surface pressure.
Q.17	(B)	Water vapour is the most abundant natural greenhouse gas, accounting for the largest share of natural infrared absorption.
Q.18	(B)	Albedo is the reflectivity of a surface; fresh snow has a very high albedo (~80–90%), while open ocean water has a low albedo.
Q.19	(B)	El Niño refers to anomalous eastern Pacific warming, often associated with deficit monsoon rainfall in India.
Q.20	(B)	Thermal inversion reverses the normal lapse rate, creating a warm lid that halts vertical convective dispersal.
Q.21	(B)	The Montreal Protocol (1987) successfully mandated global phase-outs of ozone-destroying substances.
Q.22	(B)	Proximity to the ocean moderates coastal climates, preventing extreme diurnal and annual temperature swings.
Q.23	(B)	Jet streams are high-velocity westerly or easterly geostrophic wind ribbons in the upper troposphere.
Q.24	(B)	Pre-monsoon thunderstorms in Kerala and Karnataka are popularly called Mango Showers as they hasten mango ripening.
Q.25	(A)	The carbon footprint measures total equivalent carbon dioxide emissions generated by an activity, organization, or lifestyle.